Arron Leilion

Vilnius, Lithuania • +370 665 14109 • a.leilion@euroswarms.eu
github.com/IlumCI • linkedin.com/in/arron-leilion • euroswarms.eu

Systems and AI engineer. A year on multi-agent orchestration at Swarms Corporation, where routing and orchestration work cut inference spend by 89% and took workflow execution to 118× its previous speed; since then, a ring-0 operating system in Rust with a language model running inside the kernel and every layer under it written from scratch. Equally at home in model internals and in the machine underneath them.

Experience

AI/ML Engineer, Swarms Corporation Aug 2025 – Jul 2026
Remote

- Cut large-scale API and inference spend by 89%, through routing optimisation, an orchestration redesign, and efficiency work at the infrastructure level.
- Took multi-agent workflow execution to 118× its previous speed while bringing running cost to 21% of the original baseline.
- Designed reasoning architectures and orchestration methods for multi-agent systems, and landed them in the shipping framework rather than leaving them as prototypes.
- Turned arXiv papers into working systems and internal tooling.

Independent systems and AI research Aug 2026 – present
Vilnius

Full time on the work below: an operating system, a local inference runtime, and CPU inference for models too large to hold in memory at once.

Selected work

GLaDOS github.com/IlumCI/GLaDOS
Ring-0 operating system in Rust for one specific laptop. No Unix inheritance, no syscalls, no user/kernel split: a tool call from the model is a function call. Qwen3-0.6B runs int8 in kernel space. Everything beneath it is written from scratch as well: TLS 1.3, X25519, ChaCha20, ECDSA over P-256, a TCP/IP stack, NVMe, a content-addressed store, and a composited window manager. The cryptography is checked against published RFC vectors on every boot. 40,000 lines; the only borrowed code is 509 hardware constants no datasheet publishes.

RustLMHub github.com/IlumCI/RustLMHub
Inference and training engine in Rust on the premise that a model should only have to fit on the drive, not in memory: FFN weights stream from NVMe through `O_DIRECT` while attention stays resident. GGUF k-quants, dense and MoE, CPU only, nothing borrowed from PyTorch or llama.cpp. Int8 activation kernels (2.3–2.55× on matmul), speculative decoding off the model's own multi-token head, and activation sparsity that skips up to 27.6% of neurons only where the result is provably unchanged. 267 tests, differential against reference implementations. Fine-tunes a 27B model on 16 GB of RAM.

kimi-k3-in-rust github.com/IlumCI/kimi-k3-in-rust
Runs Kimi K3, 2.78 trillion parameters and 1.56 TB of weights, on one CPU with 8.24 GB of RAM. Routing leaves 96.3% of experts dormant per token, so they live on disk behind an LRU cache instead of in memory. Output is byte-identical from 8 GB to 224 GB; only the seconds per token move. Portable C99, no BLAS, no GPU.

CR-CA and Stable Cognition github.com/IlumCI
Research frameworks: causal inference with counterfactual analysis for autonomous agents, and a long-context reasoning architecture built to be extended rather than retrained.

138 public repositories; 387 merged pull requests across open-source and commercial work.

Technical skills

Languages: Rust, Python, C, C++, Go, TypeScript, SQL

Models: transformer internals (attention geometry, RoPE, RMSNorm, tokenizers), int8 quantisation, KV-cache budgeting, constrained decoding, inference optimisation, multi-agent orchestration, PyTorch

Systems: `no_std` and UEFI, x86-64, memory management, TCP/IP, applied cryptography, NVMe and PCIe, Linux, FastAPI, distributed infrastructure

Education

Vilnius "Vyčio" Gymnasium Vilnius, Lithuania
Secondary education completed.